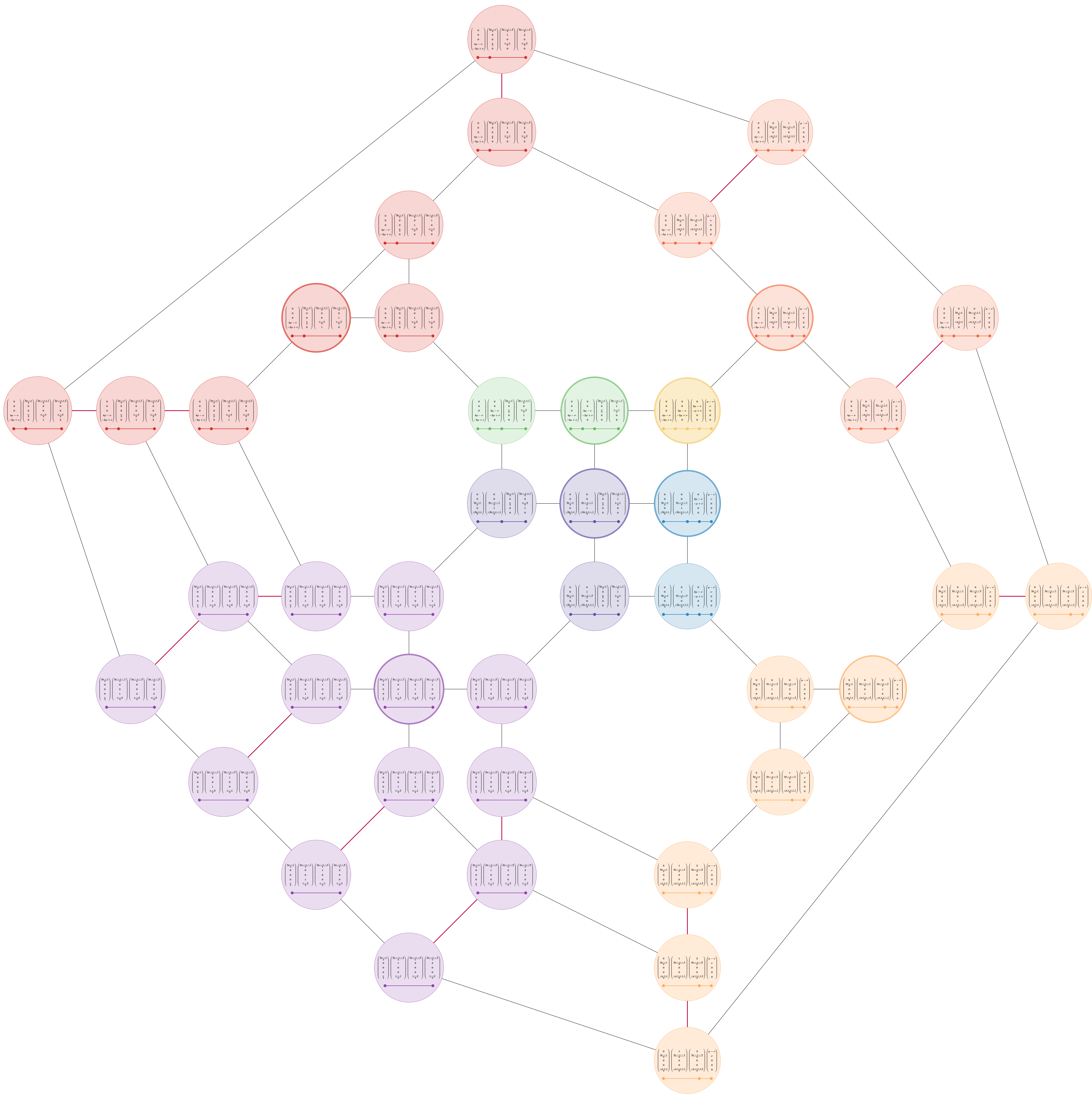




Schlegel diagram of the state polytope of the rational normal curve of degree 4. Each of the 42 vertices represents a generic weight vector and initial ideal of its toric ideal I_A . The 8 colors represent the regular subdivisions of $[0, 1, 2, 3, 4]$. Cohen–Macaulay initial ideals have double boundary. Initial ideals with the same top component are connected by a red edges.

Each vertex holds the 4 starting exponents of a basis of series solutions to the associated GKZ system $H_A(\beta)$ of linear differential operators with parameters $\beta = (\mu, \nu)$. Its solutions are Euler integrals of the form:

$$\mathcal{I}(x) = \int_{\Gamma} (x_0 + x_1 t + x_2 t^2 + x_3 t^3 + x_4 t^4)^{\mu} t^{-\nu-1} dt$$